ParaliSetu — Business Model Canvas

AI-Driven Hyperlocal Stubble Supply Chain, Baler Fleet Logistics & Escrow Marketplace

Hackathon: Hack for Social | **Theme:** Environment & Sustainability
Repository: github.com/Rishabhbansal005/Parali-setu | **Target:** Punjab & Haryana

8. KEY PARTNERS
• **Industrial Offtakers:** Bio-CNG (SATAT scheme, e.g. Verbio Sangrur), 2G ethanol (IOCL Panipat), & biomass boilers.
• **Custom Hiring Centres (CHCs):** Subsidized machinery pools with idle balers/rakes.
• **Baler & Tractor Operators:** Rural transport entrepreneurs.
• **Village Dharamkanta:** Neutral independent weighbridges.
• **Institutions:** Punjab & Haryana Agri Depts, KVKs, CAQM.

6. KEY ACTIVITIES
• **PostGIS Spatial Clustering:** Grouping smallholdings into contiguous 35-acre runs.
• **Vernacular Voice AI:** Punjabi/Hindi speech-to-booking pipeline.
• **Weighbridge Escrow Integration:** Scale ticket verification & automated T+1 ACH/UPI bank payouts.

7. KEY RESOURCES
• **Proprietary Tech:** PostGIS spatial engine & Indic speech model.
• **PAU Agronomic Formula:** Calibrated yield factors (2.0 t/acre baler recovery).
• **Ecosystem Network:** 500+ baler fleet directory & SATAT off-taker contracts.

2. VALUE PROPOSITIONS
• **Smallholder Farmers (1–4 Acres):**
- **Free field clearing** in 48–72h before wheat sowing.
- **■1,200–■1,800/acre net income** (■650/tonne direct payout).
- Zero CAQM pollution fines + In-App No-Burn Certificate.

• **Baler & Tractor Aggregators:**
- Contiguous 30-acre runs eliminate transit deadhead miles.
- **+32% diesel fuel efficiency** and maximized seasonal revenue.

• **Industrial Biomass Buyers:**
- Predictable, moisture-tested daily feedstock schedule.
- Eliminates costly digester & boiler shutdowns.

4. CUSTOMER RELATIONSHIPS
• **Voice-First Simplicity:** Zero learning curve for illiterate farmers.
• **Dharamkanta Scale Transparency:** Weight verified locally before dispatch.
• **SMS & WhatsApp Slips:** Instant payment receipts for basic phone users.

3. CHANNELS
• **Flutter Mobile App:** Lightweight 17MB Android app with offline voice fallback.
• **Next.js Operations Web Portal:** Scale ticket logging & buyer command center.
• **Village Physical Nodes:** Custom Hiring Centres (CHCs) & Dharamkanta kiosks.

1. CUSTOMER SEGMENTS
• **Primary Beneficiaries:** Small & marginal farmers (1–4 acres) facing 15-day harvest deadline. **Pays ■0.**

• **Paying Customers:** Bio-CNG (CBG) refineries under **SATAT mandatory blending (1% to 5%)**, 2G ethanol plants, and industrial boilers.

• **Supply Operators:** Custom Hiring Centre (CHC) baler owners & rural tractor fleets.

9. COST STRUCTURE
• **Technology & Platform (Fixed):** Supabase PostGIS cloud DB, Next.js hosting: ■35,000/mo. Voice Whisper/Gemini ASR tokens: ■0.27/intake.
• **Field Operations (Variable Peak):** Village Cluster Coordinators (Kisan Mitras) across 15 pilot clusters. Handheld moisture testers (■2,500/unit).
• **Compliance & Audits:** Sentinel-2 satellite processing overhead & banking escrow transaction fees.

5. REVENUE STREAMS & FINANCIAL MODEL
• **Primary B2B Take-Rate:** ■150 / Tonne (5.8% commission on ■2,600 delivered gate price paid by buyers; **0% farmer fee**).
• **Baler Optimization SaaS:** ■40 / Tonne software fee for pre-clustered contiguous runs.
• **Enterprise ESG Data:** Traceability & Scope 3 carbon audit fees for Oil Marketing Companies.
• **Financial Viability:** Break-even at 20,000 Tonnes (~10,000 acres). **Year 3 Target: ■15.0 Cr Revenue | ■12.6 Cr EBITDA.**